

# ERDŐS SIMILARITY FOR DISTANCE SETS: FINITE PATTERNS, A PINNED OBSTRUCTION, AND AN OPEN WINDOW

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ABSTRACT. For a compact set  $E \subset \mathbb{R}^d$ ,  $d \geq 2$ , let  $\Delta(E) = \{|x - y| : x, y \in E\}$  be its distance set. We ask whether distance sets of sets of large Hausdorff dimension must contain a nontrivial affine copy  $x + s\{2^{-n} : n \geq 1\}$  of the dyadic sequence. This is a distance-set version of the Erdős similarity problem. We combine two recent results announced by OpenAI: the Falconer distance theorem in all dimensions, and the dyadic case of the Erdős similarity conjecture. Assuming these, we show the following. (A) If  $\dim_{\text{H}} E > d/2$ , then  $\Delta(E)$  contains affine copies of every finite set, for a positive-measure family of translations at every small scale. (B) There is a compact set  $K \subset \mathbb{R}^d$  of nearly full volume in the unit ball whose distance set from the origin contains no affine copy of the dyadic sequence. The full distance set  $\Delta(K)$  does contain one, and so does the distance set from every other point. (C) A negative answer to the question forces  $\Delta(E)$  to have positive measure but empty interior; with the classical Mattila–Sjölin theorem, the question is therefore open exactly for  $d/2 < \dim_{\text{H}} E \leq (d+1)/2$ . Every statement of this note is proved in Lean 4 with Mathlib. The two OpenAI results enter as explicit hypotheses, and we re-checked OpenAI’s own Lean proofs of them with the Lean kernel, so Theorems A and B and Proposition C are machine-checked down to the standard axioms. The Mattila–Sjölin theorem, which is not formalized, is used only in one clearly marked remark.

## 1. INTRODUCTION

**1.1. Two problems about large sets.** A set  $P \subset \mathbb{R}$  is *measure universal* if every measurable  $S \subset \mathbb{R}$  of positive Lebesgue measure contains a nontrivial affine copy  $x + sP$  ( $x \in \mathbb{R}$ ,  $s \neq 0$ ). Steinhaus’ theorem [16] implies that every finite set is measure universal. Erdős conjectured [3] that no infinite set is. The conjecture is known for slowly decaying sequences [5] and for many structured sets [1, 9]. The geometric sequences  $\{q^n\}$ , and above all the dyadic sequence

$$\mathcal{D} = \{2^{-n} : n \geq 1\},$$

were long described as the main bottleneck [8]. In September and October 2026, OpenAI released a collection of manuscripts produced by an internal AI model. It includes a proof that  $\mathcal{D}$  is not measure universal [12] and an extension to every ratio  $q \in (0, 1)$  [14].

Falconer’s distance problem [6] asks for the smallest  $\alpha$  such that every compact  $E \subset \mathbb{R}^d$  with  $\dim_{\text{H}} E > \alpha$  has a distance set

$$\Delta(E) = \{|x - y| : x, y \in E\}$$

of positive Lebesgue measure. Falconer showed that  $d/2$  would be optimal and conjectured it. Partial progress includes [18, 4, 7]. The same OpenAI release contains a proof of the full conjecture in every dimension  $d \geq 2$  [13]. Mattila and Sjölin [11] showed that  $\dim_{\text{H}} E > (d+1)/2$  even forces  $\Delta(E)$  to contain an interval.

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**1.2. The question.** The Falconer theorem says that  $\Delta(E)$  is large in measure, while the dyadic theorem says that being large in measure is not enough to contain  $\mathcal{D}$ . Distance sets, however, have additional multi-scale structure:  $\Delta(E) \supseteq \Delta(E \cap B)$  for every ball  $B$ . In particular (Lemma 3.3), if  $\dim_{\mathbb{H}} E > d/2$  then  $\Delta(E) \cap [0, r]$  has positive measure for every  $r > 0$ . This motivates:

**Question A** (Erdős similarity for distance sets). Let  $d \geq 2$  and let  $E \subset \mathbb{R}^d$  be compact with  $\dim_{\mathbb{H}} E > d/2$ . Must  $\Delta(E)$  contain a nontrivial affine copy of  $\mathcal{D}$ ?

We could not find this question, or any study of Erdős-type patterns inside distance sets, in the literature; it is not discussed in the recent survey [8].

**1.3. Results.** Theorem A assumes the Falconer theorem (Theorem 2.1), and Theorem B assumes the dyadic avoidance theorem (Theorem 2.2); both inputs are stated precisely in Section 2.

**Theorem A** (Finite patterns). Assume Theorem 2.1. Let  $d \geq 2$ , let  $E \subset \mathbb{R}^d$  be compact with  $\dim_{\mathbb{H}} E > d/2$ , and let  $F \subset \mathbb{R}$  be finite. There is  $\delta > 0$  such that for every  $|s| < \delta$ ,

$$\mathcal{L}^1(\{x \in \mathbb{R} : x + sF \subseteq \Delta(E)\}) > 0.$$

In particular  $\Delta(E)$  contains affine copies  $x + sF$  with  $0 < s < \delta'$  and with  $-\delta' < s < 0$ , for every  $\delta' > 0$ .

**Theorem B** (A pinned obstruction). Assume Theorem 2.2. Let  $d \geq 1$  and  $\eta \in (0, 1)$ . There is a compact set  $K$  in the closed unit ball of  $\mathbb{R}^d$  with  $0 \in K$  and

$$\mathcal{L}^d(K) \geq (1 - \eta)^d \mathcal{L}^d(B(0, 1)) > 0, \quad \text{hence } \dim_{\mathbb{H}} K = d,$$

such that the pinned distance set  $\Delta_0(K) = \{|y| : y \in K\}$  contains no nontrivial affine copy of  $\mathcal{D}$ , whereas the full distance set  $\Delta(K)$  contains one. If  $d \geq 2$ , then for every  $p \in \mathbb{R}^d \setminus \{0\}$  the pinned distance set  $\Delta_p(K) = \{|p - y| : y \in K\}$  contains a nondegenerate interval, hence an affine copy of  $\mathcal{D}$ .

So pinned distance sets are not universal hosts for  $\mathcal{D}$ , even from a pin inside a set of full dimension and nearly full volume in the ball. The obstruction, however, sits at one single point. Theorem B holds verbatim for every ratio  $q \in (0, 1)$  in place of  $1/2$ , assuming the corresponding avoidance statement (Theorem 4.6).

**Proposition C.** Let  $d \geq 2$ , let  $E \subset \mathbb{R}^d$  be compact, and let  $q \in (0, 1)$ .

- (i) If  $\Delta(E)$  has nonempty interior, it contains an affine copy of every bounded set  $P \subseteq [0, M]$  and of  $\{q^n : n \geq 1\}$ .
- (ii) This applies whenever  $\mathcal{L}^d(E) > 0$ .
- (iii) Assume Theorem 2.1. If  $\dim_{\mathbb{H}} E > d/2$  and  $\Delta(E)$  contains no affine copy of  $\{q^n : n \geq 1\}$ , then  $\mathcal{L}^1(\Delta(E)) > 0$ ,  $\Delta(E)$  has empty interior, and  $\mathcal{L}^d(E) = 0$ .

Combined with the Mattila–Sjölin theorem, a counterexample to Question A must have  $d/2 < \dim_{\mathbb{H}} E \leq (d + 1)/2$  (Remark 5.1), so the question is open exactly in this window. We are not aware of a general result that guarantees nonempty interior of  $\Delta(E)$  below the Mattila–Sjölin threshold  $(d + 1)/2$  (see [10] for background). Under that reading, a negative answer to Question A would also give new examples of distance sets with positive measure and empty interior. Section 6 lists further questions.

**1.4. Formalization and status of the inputs.** *Every numbered statement of this note is proved in Lean 4 [2] with Mathlib [17].* The Lean name is given next to each statement and in Table 1. External results enter only as explicit hypotheses, never as axioms:

- (1) the Falconer theorem [13] (Theorem 2.1), used in Theorem A, Lemma 3.3, Proposition 3.2 and Proposition C(iii);
- (2) the dyadic avoidance theorem [12] (Theorem 2.2), used in Theorem B;
- (3) the geometric avoidance statement for a ratio  $q \neq 1/2$  [14] (Statement 2.3), used only in Theorem 4.6;
- (4) the Mattila–Sjölin theorem [11] (Statement 2.4), used only in Remark 5.1.

Inputs (1) and (2) come from an AI-generated, unrefereed manuscript collection whose authors state that some results “could have issues”. OpenAI also provides Lean proofs of both. We rebuilt the relevant part of that library ourselves and checked with the Lean kernel that these proofs establish exactly our hypotheses (Section 7). Consequently Theorems A and B, Proposition C and all lemmas are machine-checked down to Lean’s standard axioms. Input (3) has no Lean proof, and input (4) is a classical peer-reviewed theorem that has not been formalized. A formal proof certifies the statement as formalized; it does not replace independent mathematical review of [13, 12].

## 2. NOTATION AND INPUTS

$\mathcal{L}^d$  is Lebesgue measure on  $\mathbb{R}^d$ ,  $|\cdot|$  the Euclidean norm,  $B(0, 1)$  the open unit ball, and  $\dim_{\text{H}}$  Hausdorff dimension. For  $E \subset \mathbb{R}^d$  and  $p \in \mathbb{R}^d$  we write  $\Delta(E) = \{|x - y| : x, y \in E\}$  and  $\Delta_p(E) = \{|p - y| : y \in E\}$ . An *affine copy* of  $P \subset \mathbb{R}$  is a set  $x + sP$  with  $x \in \mathbb{R}$  and  $s \neq 0$ . For  $q \in (0, 1)$  let  $\mathcal{G}_q = \{q^n : n \geq 1\}$ , so  $\mathcal{D} = \mathcal{G}_{1/2}$ .

**Theorem 2.1** (Falconer distance theorem [13]). *Let  $d \geq 2$  and let  $E \subset \mathbb{R}^d$  be compact with  $\dim_{\text{H}} E > d/2$ . Then  $\mathcal{L}^1(\Delta(E)) > 0$ .* [Lean: FalconerStatement]

**Theorem 2.2** (Dyadic avoidance [12]). *For every  $\eta \in (0, 1)$  there is a compact set  $A \subseteq [0, 1]$  with  $\mathcal{L}^1(A) > 1 - \eta$  such that for every  $x \in \mathbb{R}$  and every  $s \neq 0$  there is  $n \geq 1$  with  $x + s2^{-n} \notin A$ .* [Lean: DyadicAvoidanceStatement]

**Statement 2.3** (Geometric avoidance for the ratio  $q$ ). *Statement GA( $q$ ): Theorem 2.2 with  $2^{-n}$  replaced by  $q^n$ . For  $q = 1/2$  it is Theorem 2.2; for all  $q \in (0, 1)$  it is claimed in [14], without a Lean proof.* [Lean: GeometricAvoidanceStatement]

**Statement 2.4** (Mattila–Sjölin [11]). *If  $d \geq 2$  and  $E \subset \mathbb{R}^d$  is compact with  $\dim_{\text{H}} E > (d+1)/2$ , then  $\Delta(E)$  has nonempty interior.* [Lean: MattilaSjolinStatement]

## 3. FINITE PATTERNS

The following is the standard Steinhaus-type argument (compare the introduction of [12]). We include it because Theorem A needs the version with a positive-measure set of translations at every small scale.

**Lemma 3.1.** *Let  $S \subset \mathbb{R}$  be measurable with  $0 < \mathcal{L}^1(S) < \infty$  and let  $F \subset \mathbb{R}$  be finite. There is  $\delta > 0$  such that  $\mathcal{L}^1(\{x : x + sF \subseteq S\}) > 0$  for all  $|s| < \delta$ .* [Lean: eventually\_volume\_pattern\_pos]

*Proof.* For  $a \in F$  let  $T_{a,s} = S - sa$ . Translation is continuous in measure:  $\mathcal{L}^1(T_{a,s} \triangle S) \rightarrow 0$  as  $s \rightarrow 0$ . Choose  $\delta > 0$  with  $\sum_{a \in F} \mathcal{L}^1(T_{a,s} \triangle S) < \mathcal{L}^1(S)$  for  $|s| < \delta$ . Since

$$S \subseteq \{x : x + sF \subseteq S\} \cup \bigcup_{a \in F} (S \setminus T_{a,s}),$$

we get  $\mathcal{L}^1(S) \leq \mathcal{L}^1(\{x : x + sF \subseteq S\}) + \sum_{a \in F} \mathcal{L}^1(T_{a,s} \Delta S)$ , and the claim follows.  $\square$

*Proof of Theorem A.*

[Lean: finite\_pattern\_distSet, exists\_finite\_pattern\_distSet]

$\Delta(E)$  is the image of the compact set  $E \times E$  under the continuous map  $(x, y) \mapsto |x - y|$ , so it is compact. In particular it is measurable with finite measure, and  $\mathcal{L}^1(\Delta(E)) > 0$  by Theorem 2.1. Lemma 3.1 with  $S = \Delta(E)$  gives the first claim. For the second, apply it with  $s = \pm \min(\delta, \delta')/2$ ; a set of positive measure is nonempty.  $\square$

**Proposition 3.2** (Planar form). *Under the assumptions of Theorem A, the set  $\{(s, x) \in \mathbb{R}^2 : x + sF \subseteq \Delta(E)\}$  has positive planar Lebesgue measure.* [Lean: volume\_pattern\_pairs\_pos]

*Proof.* The set is closed, hence measurable. By Tonelli's theorem its measure is  $\int \mathcal{L}^1(\{x : x + sF \subseteq \Delta(E)\}) ds$ , and the integrand is positive for  $|s| < \delta$ .  $\square$

**Lemma 3.3** (Distances at every scale). *Assume Theorem 2.1. If  $d \geq 2$  and  $E \subset \mathbb{R}^d$  is compact with  $\dim_{\mathbb{H}} E > d/2$ , then  $\mathcal{L}^1(\Delta(E) \cap [0, r]) > 0$  for every  $r > 0$ .* [Lean: volume\_distSet\_inter\_Icc\_pos]

*Proof.* Cover  $E$  by finitely many closed balls  $\bar{B}(c_i, r/2)$ . Hausdorff dimension of a finite union is the maximum of the dimensions, so some compact piece  $E_i = E \cap \bar{B}(c_i, r/2)$  has  $\dim_{\mathbb{H}} E_i > d/2$ . By Theorem 2.1,  $\mathcal{L}^1(\Delta(E_i)) > 0$ , and  $\Delta(E_i) \subseteq \Delta(E) \cap [0, r]$ .  $\square$

*Remark 3.4.* Theorem A uses nothing about distance sets beyond compactness and positive measure. Its point is the contrast with Theorem B and Question A: the finite-to-infinite gap of the Erdős similarity problem reappears inside Falconer distance sets.

#### 4. THE PINNED OBSTRUCTION

**Lemma 4.1** (Adding the limit point). *Let  $q \in (0, 1)$  and let  $A \subset \mathbb{R}$  contain no nontrivial affine copy of  $\mathcal{G}_q$ . Then neither does  $A \cup \{0\}$ .* [Lean: not\_containsGeomCopy\_insert\_zero]

*Proof.* Suppose  $x + sq^n \in A \cup \{0\}$  for all  $n \geq 1$ , with  $s \neq 0$ . The points  $x + sq^m$ ,  $m \geq 0$ , are pairwise distinct. If none of them is 0, then  $x + s\mathcal{G}_q \subseteq A$ , a contradiction. Otherwise  $x + sq^m = 0$  for exactly one  $m \geq 0$ . The tail  $\{x + sq^{m+n} : n \geq 1\} = x + (sq^m)\mathcal{G}_q$  is then an affine copy of  $\mathcal{G}_q$  avoiding 0, hence contained in  $A$ , again a contradiction.  $\square$

**Lemma 4.2** (Radial sets). *Let  $d \geq 1$  and let  $A \subseteq [0, 1]$  be compact with  $\mathcal{L}^1(A) = c$ . Then  $K_A = \{y \in \mathbb{R}^d : |y| \in A\}$  satisfies  $\mathcal{L}^d(K_A) \geq c^d \mathcal{L}^d(B(0, 1))$ .* [Lean: volume\_norm\_preimage\_ge]

*Proof.* In polar coordinates,

$$\mathcal{L}^d(K_A) = d \mathcal{L}^d(B(0, 1)) \int_A r^{d-1} dr.$$

Since  $r \mapsto r^{d-1}$  is nondecreasing,

$$\begin{aligned} \int_A r^{d-1} dr - \int_0^c r^{d-1} dr &= \int_{A \setminus [0, c]} r^{d-1} dr - \int_{[0, c] \setminus A} r^{d-1} dr \\ &\geq c^{d-1} (\mathcal{L}^1(A \setminus [0, c]) - \mathcal{L}^1([0, c] \setminus A)) = c^{d-1} (\mathcal{L}^1(A) - c) = 0, \end{aligned}$$

so  $\int_A r^{d-1} dr \geq c^d/d$ .  $\square$

**Lemma 4.3.** *If  $K \subset \mathbb{R}^d$  has  $\mathcal{L}^d(K) > 0$ , then  $\dim_{\mathbb{H}} K = d$ .* [Lean: dimH\_eq\_of\_volume\_pos]

*Proof.* For the sup-norm on  $\mathbb{R}^d$ ,  $d$ -dimensional Hausdorff measure equals  $\mathcal{L}^d$ , and the Euclidean and sup norms are bi-Lipschitz equivalent. Hence  $\mathcal{H}^d(K) > 0$  and  $\dim_{\mathbb{H}} K \geq d$ . The reverse inequality is trivial.  $\square$

**Lemma 4.4** (Steinhaus). *If  $K \subset \mathbb{R}^d$  ( $d \geq 1$ ) is compact with  $\mathcal{L}^d(K) > 0$ , then  $[0, \varepsilon] \subseteq \Delta(K)$  for some  $\varepsilon > 0$ .* [Lean: exists\_Ico\_subset\_distSet]

*Proof.* By Steinhaus' theorem  $K - K \supseteq B(0, \varepsilon)$  for some  $\varepsilon > 0$ . For  $0 \leq r < \varepsilon$  write  $re_1 = x - y$  with  $x, y \in K$ ; then  $r = |x - y| \in \Delta(K)$ . □

**Lemma 4.5** (Other pins). *Let  $d \geq 2$ ,  $\rho > 0$ ,  $p \in \mathbb{R}^d \setminus \{0\}$ , and let  $K \subseteq \mathbb{R}^d$  contain the sphere  $\{|y| = \rho\}$ . Then  $\Delta_p(K) \supseteq [|p| - \rho, |p| + \rho]$ .* [Lean: Icc\_subset\_pinnedDistSet]

*Proof.* The sphere is connected because  $d \geq 2$ , and the continuous function  $y \mapsto |p - y|$  on it takes the values  $||p| - \rho|$  at  $\rho p/|p|$  and  $|p| + \rho$  at  $-\rho p/|p|$ . By the intermediate value theorem it takes every value in between. □

**Theorem 4.6** (Theorem B for a general ratio). *Let  $q \in (0, 1)$  and assume  $\text{GA}(q)$  (Statement 2.3). Then Theorem B holds with  $\mathcal{D}$  replaced by  $\mathcal{G}_q$ .* [Lean: pinned\_geometric\_obstruction]

*Proof.* Let  $A$  be given by  $\text{GA}(q)$  for  $\eta$ , and put  $K = \{y : |y| \in A \cup \{0\}\}$ . The set  $A \cup \{0\} \subseteq [0, 1]$  is compact, so  $K$  is compact and lies in the closed unit ball, and  $0 \in K$ . Since  $K \supseteq K_A$  and  $\mathcal{L}^1(A) > 1 - \eta$ , Lemma 4.2 gives  $\mathcal{L}^d(K) \geq (1 - \eta)^d \mathcal{L}^d(B(0, 1)) > 0$ , and Lemma 4.3 gives  $\dim_{\text{H}} K = d$ . Further,  $\Delta_0(K) = \{|y| : y \in K\} \subseteq A \cup \{0\}$ , which contains no affine copy of  $\mathcal{G}_q$  by Lemma 4.1.

Any set  $P \subseteq \mathbb{R}$  containing a nondegenerate interval  $[a, b]$  contains the affine copy  $a + (b - a)\mathcal{G}_q$ . By Lemma 4.4,  $\Delta(K)$  contains an interval. For  $d \geq 2$  and  $p \neq 0$ , pick  $\rho \in A$  with  $\rho > 0$  (possible since  $\mathcal{L}^1(A) > 0$ ); the sphere  $\{|y| = \rho\}$  lies in  $K$ , so by Lemma 4.5  $\Delta_p(K)$  contains an interval of length  $2 \min(\rho, |p|) > 0$ . □

*Proof of Theorem B.* [Lean: pinned\_dyadic\_obstruction]

Theorem 4.6 with  $q = 1/2$ , where  $\text{GA}(1/2)$  is Theorem 2.2. □

## 5. WHERE THE QUESTION IS OPEN

*Proof of Proposition C.* [Lean: distSet\_containsAffineCopy\_of\_interior, distSet\_containsAffineCopy\_of\_volume\_pos, counterexample\_properties]

(i) A set with nonempty interior contains an interval  $[a, b]$  with  $a < b$ . For  $P \subseteq [0, M]$ ,  $M > 0$ , we have  $a + \frac{b-a}{M}P \subseteq [a, b]$ , and  $\mathcal{G}_q \subseteq [0, 1]$ . (ii) follows from Lemma 4.4. (iii)  $\mathcal{L}^1(\Delta(E)) > 0$  is Theorem 2.1; nonempty interior or  $\mathcal{L}^d(E) > 0$  would give a copy of  $\mathcal{G}_q$  by (i) and (ii). □

**Remark 5.1** (Uses the Mattila-Sjölin theorem). Assume Statement 2.4. If  $d \geq 2$ ,  $E \subset \mathbb{R}^d$  is compact and  $\Delta(E)$  contains no affine copy of  $\mathcal{G}_q$ , then  $\dim_{\text{H}} E \leq (d + 1)/2$ : otherwise  $\Delta(E)$  has nonempty interior and Proposition C(i) applies. Together with Proposition C(iii), a counterexample to Question A lies in the window  $d/2 < \dim_{\text{H}} E \leq (d + 1)/2$ . [Lean: window\_of\_mattilaSjolin]

## 6. OPEN QUESTIONS

- (1) Question A in the window  $d/2 < \dim_{\text{H}} E \leq (d + 1)/2$ . A positive answer would need multi-scale structure of distance sets beyond positive measure. A negative answer would need a construction controlling all distances of a set of dimension  $> d/2$  simultaneously.
- (2) *All pins.* Is there a compact  $E \subset \mathbb{R}^d$  with  $\mathcal{L}^d(E) > 0$  (or merely  $\dim_{\text{H}} E > d/2$ ) such that  $\Delta_p(E)$  contains no affine copy of  $\mathcal{D}$  for every  $p \in E$ ? By Lemma 6.1 below, this asks for a set whose complement contains, around every point of  $E$ , a sphere of radius  $r$  for some  $r$  in every affine copy of  $\mathcal{D}$ . This suggests a connection with spherical maximal function estimates. Pinned Falconer-type theorems [15, 7] give pins with large pinned distance sets, but say nothing about such patterns.

- (3) The same questions for general infinite sets  $P$ , for instance for every  $P$  known to be non-universal.

**Lemma 6.1.** *For  $p \in \mathbb{R}^d$  and  $E \subseteq \mathbb{R}^d$ , the pinned distance set  $\Delta_p(E)$  contains no affine copy of  $\mathcal{G}_q$  if and only if for all  $x \in \mathbb{R}$  and  $s \neq 0$  there is  $n \geq 1$  such that the sphere of radius  $x + sq^n$  about  $p$  is disjoint from  $E$ .* *[Lean: `not_containsGeomCopy_pinnedDistSet_iff`]*

*Proof.*  $r \in \Delta_p(E)$  if and only if the sphere of radius  $r$  about  $p$  meets  $E$ . □

## 7. FORMALIZATION

The Lean development accompanying this note is available at <https://github.com/haraldschilly/math-distance-set-similarity>. It is pinned to Lean 4 v4.34.1 and Mathlib commit d13f23b, the same versions as OpenAI’s library, so the input statements typecheck identically. The four hypotheses of Section 1.4 are Lean propositions in `Statements.lean`. `FalconerStatement` and `DyadicAvoidanceStatement` are copied verbatim from OpenAI’s statement files `FalconerAllDimensions.lean` and `DyadicAvoidance.lean`. Table 1 lists every result of this note with its Lean name and its hypotheses. For example, Theorem B reads:

```
theorem pinned_dyadic_obstruction (hDy : DyadicAvoidanceStatement) {d : ℕ}
  (hd : 1 ≤ d) {η : ℝ} (hη0 : 0 < η) (hη1 : η < 1) :
  ∃ K : Set (EuclideanSpace ℝ (Fin d)),
    IsCompact K ∧ K ⊆ closedBall 0 1 ∧
    (0 : EuclideanSpace ℝ (Fin d)) ∈ K ∧
    ENNReal.ofReal ((1 - η) ^ d) * volume (ball 0 1) ≤ volume K ∧
    0 < volume K ∧ dimH K = d ∧
    ¬ ContainsDyadicCopy (pinnedDistSet 0 K) ∧ ContainsDyadicCopy (distSet K) ∧
    (2 ≤ d → ∀ p, p ≠ 0 → ContainsDyadicCopy (pinnedDistSet p K))
```

| Result                | Lean name                                                              | Hypotheses       |
|-----------------------|------------------------------------------------------------------------|------------------|
| Lemma 3.3             | <code>volume_distSet_inter_Icc_pos</code>                              | Falconer         |
| Lemma 3.1             | <code>eventually_volume_pattern_pos</code>                             | none             |
| Theorem A             | <code>finite_pattern_distSet, exists_finite_pattern_distSet</code>     | Falconer         |
| Proposition 3.2       | <code>volume_pattern_pairs_pos</code>                                  | Falconer         |
| Lemma 4.1             | <code>not_containsGeomCopy_insert_zero</code>                          | none             |
| Lemma 4.2             | <code>volume_norm_preimage_ge</code>                                   | none             |
| Lemma 4.3             | <code>dimH_eq_of_volume_pos</code>                                     | none             |
| Lemma 4.4             | <code>exists_Ico_subset_distSet</code>                                 | none             |
| Lemma 4.5             | <code>Icc_subset_pinnedDistSet</code>                                  | none             |
| Theorem B             | <code>pinned_dyadic_obstruction</code>                                 | dyadic avoidance |
| Theorem 4.6           | <code>pinned_geometric_obstruction</code>                              | $\text{GA}(q)$   |
| Proposition C(i),(ii) | <code>distSet_containsAffineCopy_of_interior, ..._of_volume_pos</code> | none             |
| Proposition C(iii)    | <code>counterexample_properties</code>                                 | Falconer         |
| Remark 5.1            | <code>window_of_mattilaSjolin</code>                                   | Mattila–Sjölin   |
| Lemma 6.1             | <code>not_containsGeomCopy_pinnedDistSet_iff</code>                    | none             |

TABLE 1. Results and their Lean formalizations (namespace `DistanceSimilarity`). Every listed theorem depends only on the axioms `propext`, `Classical.choice`, `Quot.sound`.

*Re-checking the inputs.* A script in the repository copies, from a checkout of [github.com/openai/math](https://github.com/openai/math) at commit `adc7f12`, the transitive closure of OpenAI’s Lean modules proving the two inputs: 2459 files, depending only on Mathlib. These are built

with the same toolchain. A further module states `theorem : FalconerStatement := OAI.Falconer.falconer_distance_conjecture` and the analogous dyadic statement. It then derives unconditional versions of Lemma 3.3, Theorem A, Proposition 3.2, Theorem B and Proposition C(iii). The kernel accepts these, and `#print axioms` reports only the three standard axioms. We additionally replayed the entire resulting environment (Mathlib, OpenAI’s modules and ours) in a fresh kernel with Lean’s `leanchecker`. We did not run OpenAI’s `comparator` setup. A formal proof certifies the statement as formalized. Readers should still compare the formal statements with the informal ones; to make this easy, the repository contains a plain-language guide to reading the Lean statements.

## ACKNOWLEDGMENTS AND USE OF AI

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